APPENDIX A VERIFICATION OF FULL GILLESPIE ON CLIQUES

We check the Full Gillespie implementation on clique networks, for which mean infection times can be calculated analytically. A clique is a community in which every pair of nodes is connected. We use the first community of the clique chains: it has n nodes and k contacts with the next community. The k source nodes with these contacts are the bridge nodes. The initially infected node and the set of bridge nodes are selected uniformly and independently.

Two times are distinguished. The bridge infection time, T_B, is when the first bridge node becomes infected. The export time, T_{export}, is when infection is first transmitted to the next community. Infection of a bridge makes transmission possible, but transmission need not occur immediately. The last column of Table 1 measures the additional waiting time, $T_{\mathrm{export}} - T_B$, in Full Gillespie.

Table 1 compares analytical means with simulation results. For example, for $n = 100$ and $k = 1$, the simulated mean bridge infection time is 17.05 ± 0.38, compared with the exact value 17.26. The simulated mean wait from bridge infection to export is 326.5 ± 17.0, compared with the exact value 333.3. These comparisons check internal spread and, for one contact, the subsequent transmission to the next community.

Table 1. Checks on the first community of the clique chains ($\beta = 0.003$, 400 runs per configuration). T_B is the first bridge infection time; T_{export} is the first transmission time to the next community. Full values are means $\pm$ standard errors. Dashes indicate that the one-contact waiting-time check is not reported for $k = 5$.

n	k	Exact mean T_B	Full mean T_B	Full mean wait $T_{\mathrm{export}} - T_B$
50	1	29.86	29.72 ± 0.71	354.0 ± 16.9
100	1	17.26	17.05 ± 0.38	326.5 ± 17.0
250	1	8.13	7.84 ± 0.15	347.4 ± 16.6
1000	1	2.49	2.51 ± 0.04	353.8 ± 17.9
50	5	15.97	15.96 ± 0.55	–
100	5	10.31	9.64 ± 0.29	–
250	5	5.35	5.16 ± 0.12	–
1000	5	1.80	1.77 ± 0.03	–

For $k = 1$, the delay after bridge infection is an independent $\mathrm{Exp}(\beta)$ waiting time, with mean $1/\beta = 333.3$ at $\beta = 0.003$. For $k = 5$, other bridge nodes may become infected before the first export, increasing the total transmission rate. The same constant-rate waiting-time check therefore does not apply. The waiting time is still defined, but this table reports it only for $k = 1$; the dashes do not indicate missing runs.

The analytical bridge infection time follows from the SI process on a clique. With m infected and $n - m$ susceptible nodes, there are $m(n-m)$ possible transmitting contacts, each with rate β. The mean wait for the next internal infection is therefore $1/[\beta m(n-m)]$. The probability that no bridge is among the first m infected nodes is $\binom{n-k}{m}/\binom{n}{m}$. Adding the waiting times weighted by these probabilities gives

$$\mathbb{E}[T_B] = \sum_{m=1}^{n-k} \frac{\binom{n-k}{m}}{\binom{n}{m}} \frac{1}{\beta m(n-m)}, \tag{1}$$

The case in which the initial infected node is already a bridge has $T_B = 0$ and is included in this expectation. For one bridge, the expression simplifies to

$$\mathbb{E}[T_B] = \frac{H_{n-1}}{\beta n} \quad (k = 1), \tag{2}$$

where $H_{n-1} = \sum_{r=1}^{n-1} 1/r$ is the harmonic number. The mean time to infect the entire clique is $2H_{n-1}/(\beta n)$.